Towards AI-Enabled Nuclear Astrophysics

A Research Proposal Report

Lyu Shenghao[1,2]

[1]Tsinghua University, Beijing, China

[2]China Institute of Atomic Energy, Beijing, China

October 16, 2026





Table of Contents

1. **Introduction**
2. **Reference**
   - Part I: AI for Scientific Research
   - Part II: ML in Nuclear Astrophysics
3. **Proposal**
   - Approach, Roadmap and Budget

Introduction: Why AI?

Nuclear astrophysics is suitable for AI due to some characteristics.

- **Interdisciplinary with Knowledge Gaps**

  Relevant knowledge is distributed across nuclear physics, astrophysics, and observational literature.

- **Simulation-Based & Computationally Constrained**

  Full nucleosynthesis simulations may involve $\sim 10^3$ isotopes and up to $\sim 10^4$ reactions, coupling with 2D/3D hydrodynamics.

- **Experimentally Expensive**

  Experiments involve many coupled setup parameters, while beam time and machine access are limited and costly.

- **Uncertainty-Sensitivity Intensive**

  Make predictions and identify which uncertain inputs dominate the results should be prioritized for theory and experiments.

Introduction: ML in Nuc/Astro

Machine learning has already been widely explored in both nuclear physics and astrophysics.

Nuclear Physics

- **Nuclear theory**
  DFT emulation [1]
- **Calorimetry**
  GlueX [2]
- **Particle identification**
  DeepRICH [3]
- **Accelerator control**
  Real-time tuning [4]

Nuclear Astrophysics ⟷

Astrophysics

- **Cosmology**
  Moriwaki et al. [5]
- **Galaxy evolution**
  Fraser et al. [6]
- **Cosmic-ray analysis**
  Arimura [7]
- **Stellar abundances**
  den Hartogh et al. [8]

However, while ML has advanced both fields separately, its use in nuclear astrophysics remains very limited [9].

Reference

Part I: AI for Scientific Research

Robin: a closed lab-in-loop workflow

Scientific question: Find potential treatments for dry age-related macular degeneration (dAMD).

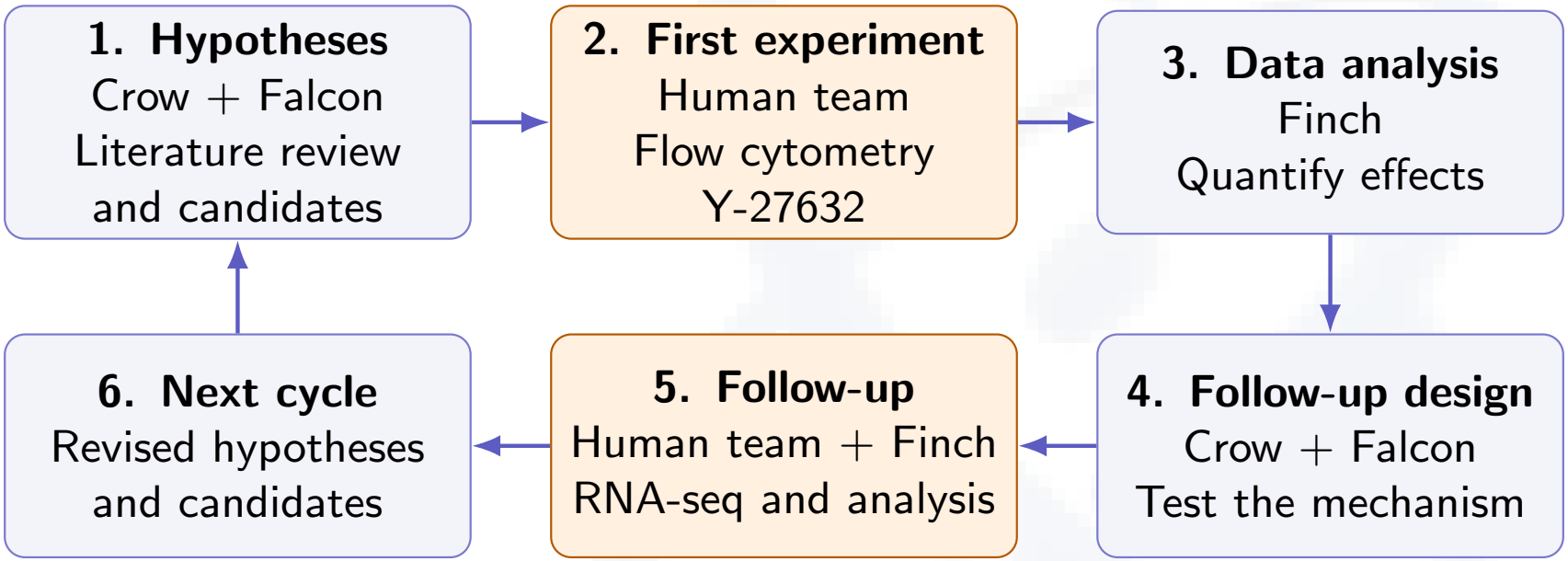


Figure 1: Robin's two-experiment research loop. Redrawn from Ghareeb et al.

Findings: Ripasudil(best) and KL001(new) enhanced phagocytosis; RNA-seq implicated ABCA1 in the ripasudil response.

Robin, Nature (2026)

Techniques

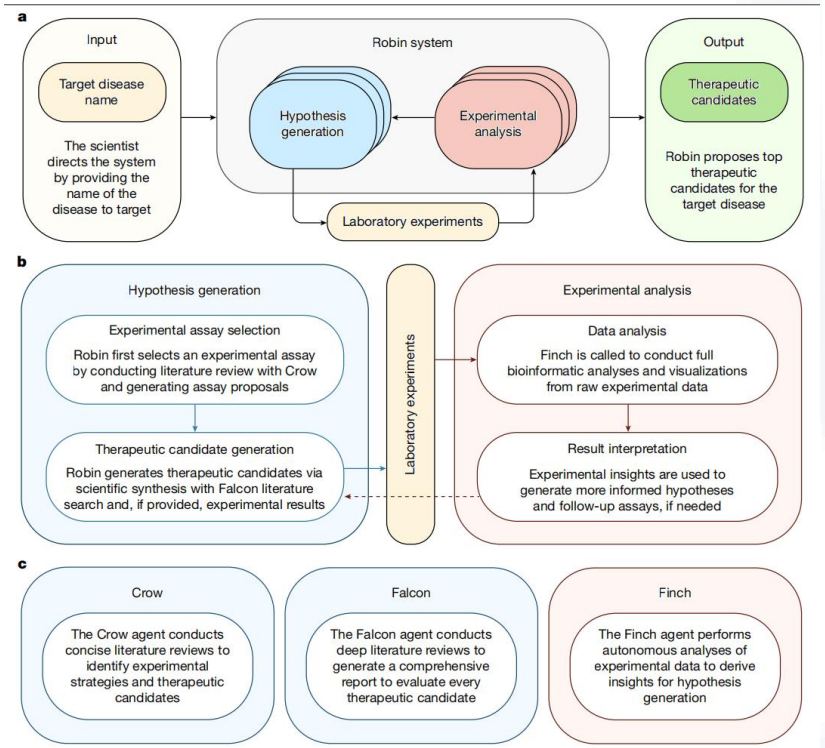


Figure 2: Robin architecture

MAS

- Crow / Falcon: PaperQA2
- Finch: Jupyter kernel

Evaluation

- Position bias in LLM judges
- Tournament
- Bradley–Terry–Luce (BTL)

Prompt engineering

Robin, Nature (2026)

Co-Scientist: focus on generating hypotheses

Three techniques

1. Self-play debate
2. Tournament-based evolution
3. Agentic capability

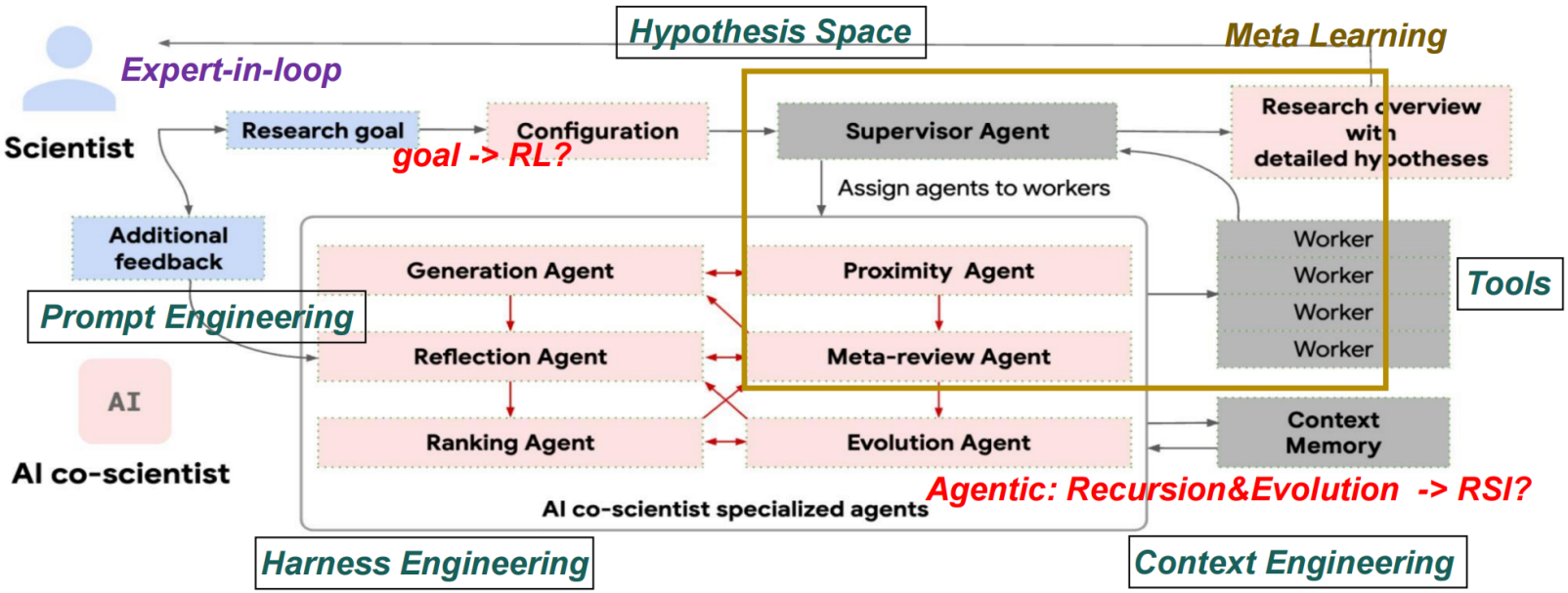


Figure 3: Co-Scientist architecture with annotated research concepts.

Towards an AI co-scientist, arXiv:2502.18864

Co-Scientist: test-time compute scaling

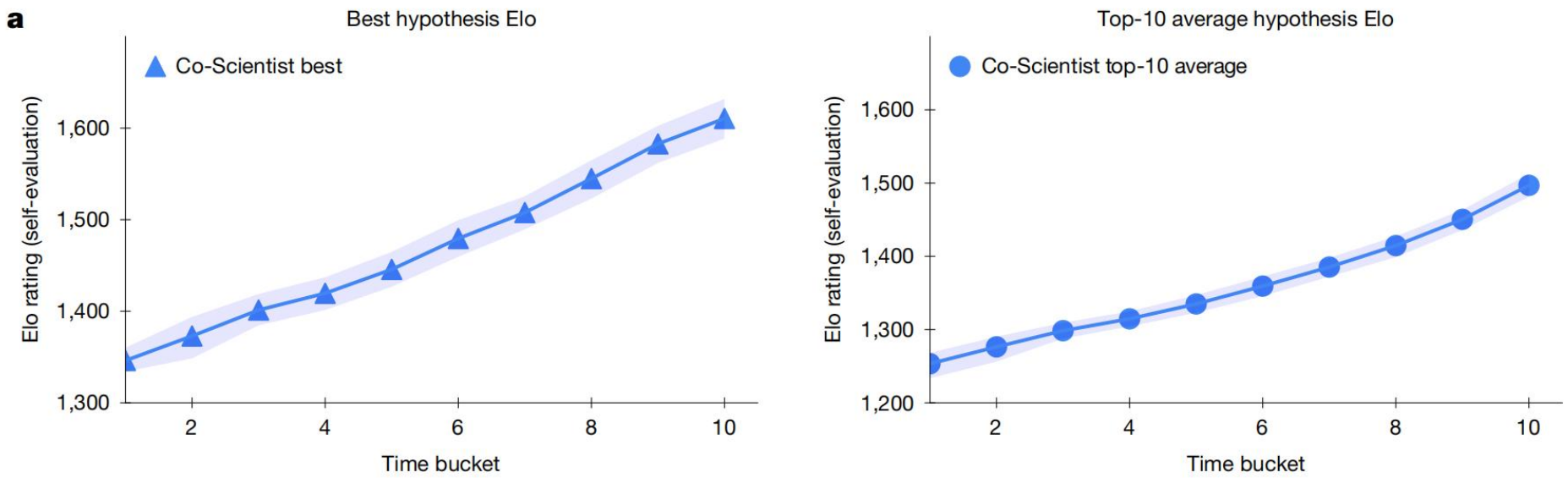


Figure 4: Internal Elo ratings of the best and top-ten hypotheses over time.

Observation: More inference-time search improves internally ranked hypotheses.

Limits

1. No saturation was observed within the tested compute range.
2. No general quantitative law in FLOPs, tokens, or cost was established.
3. Internal Elo ratings cannot replace experimental success rates.

Towards an AI co-scientist, arXiv:2502.18864

Open Question: Emergence

The paper's perspective

Context memory and iterative refinement form an increasingly coherent, interconnected representation of the scientific research process.

Meta-review carries lessons into subsequent prompts, enabling improvement without updating model parameters.

Open question

Can scientific intelligence emerge from this evolving internal model of the research process?

Co-Scientist, Nature (2026): Discussion and Methods

Co-Scientist: focus on generating hypotheses

Default Criteria

1. **Alignment** with the research goal.
2. **Plausibility:** Check reasoning and scientific assumptions.
3. **Novelty:** Go beyond existing knowledge; explain mechanisms.
4. **Testability:** Validate empirically through experiments.
5. **Safety:** Control outputs to prevent unsafe or harmful research.

Additional criteria can be configured.

Towards an AI co-scientist, arXiv:2502.18864

Scientific Exploration

Search + Optimization

A **combinatorial search and optimization problem** in hypothesis space

Exploration $\longleftrightarrow$ Exploitation

Balance finite compute between new ideas and refinement.

Towards an AI co-scientist, arXiv:2502.18864

New techniques to be explored...

Technique	**Current limitation in Co-Scientist**
World model	Context-based reasoning; no explicit executable scientific world model.
Reinforcement learning	Research feedback does not update policy parameters online.
Bayesian design	No complete information-gain-driven experimental design loop.
Formal verification	LLM review rather than mathematical or physical proofs.
Causal discovery	Mechanistic proposals without a unified causal-discovery algorithm.
Multi-objective search	Several criteria; candidates mainly organized by tournaments.
Continual learning	Persistent context rather than continual parameter learning.
Closed-loop experiments	Real validation, without fully automated online learning.

Towards an AI co-scientist, arXiv:2502.18864

XunZi: multi-model data fusion

Motivation: Reduce hallucinations with multimodal empirical data.

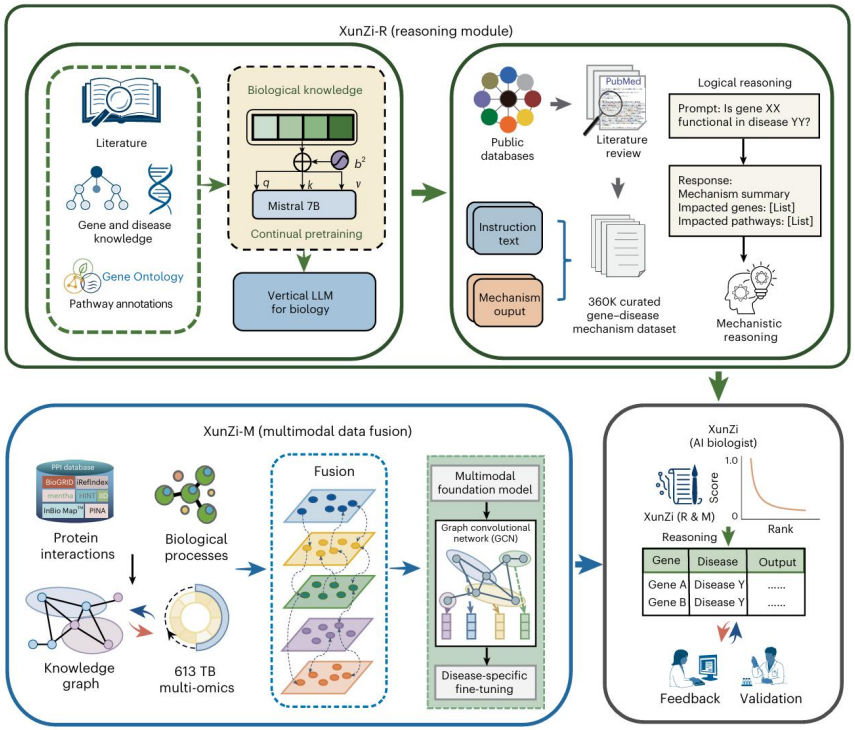


Figure 5: XunZi architecture

XunZi-R

Reasoning model

Continual pretraining

+ supervised fine-tuning

XunZi-M

GCN fusion model

Dataset

1. Publications
2. Structured biological data
3. CoT

   Manually checked

XunZi, Nature Biomedical Engineering (2026)

XunZi: multi-model data fusion

XunZi-R: Mistral-7B

- **CPT:** 24.41M PubMed abstracts + 2.05M structured knowledge records; next-token prediction.

$$\mathcal{L}_{\mathrm{CLM}} = -\sum_{t=1}^{T} \log P(x_t \mid x_{<t}; \theta)$$

- **SFT:** 330,000 gene–disease mechanism examples.

$$\mathcal{L}_{\mathrm{SFT}} = -\sum_{t=1}^{T} \log P(y_t \mid y_{<t}, x; \theta)$$

- **Output:** Confidence s for feature fusion, plus a mechanistic explanation for interpretability.

XunZi, Nature Biomedical Engineering (2026)

XunZi: multi-model data fusion

XunZi-M: Two-layer GCN

- **Data:** 2.81M PPI records + 47,900 GO terms; 621,000 nodes and 6.09M edges.
- Predict disease-specific gene relevance with binary cross-entropy.

$$\mathcal{L}_{\mathrm{BCE}} = -\sum_{i \in \mathcal{Y}} [y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)]$$

- **Output:** Continuous scores rank candidate targets.
- **Fusion:** Concatenate graph features with s; a linear layer produces the final relevance score.

XunZi, Nature Biomedical Engineering (2026)

IdeaScientist: RL in hypothesis generation

- **Insight:** Transfer mechanisms that solved analogous challenges in other fields.
- **Three roles:** Gap finding, innovation, and report writing; each trained with reinforcement learning.
- **Corpus:** Svalbard Idea Vault, 2.77M decomposed research ideas.
- **Evaluation:** Train before a temporal cutoff; test after it.

Natural intersection

Particle physics, nuclear physics, astrophysics, cosmology, and AI.

IdeaScientist, arXiv:2610.04074 (2026)

IdeaScientist: RL in hypothesis generation

The Svalbard Idea Vault

- An offline corpus of **2.77 million decomposed ideas**.
- Each tuple contains problem, challenge, gap, intuition, and proposal.
- Papers from 1986 to June 18, 2026; reference checks and topic-relevant method filtering.
- **Inputs:** Problem definition and challenge.
- **Three targets:** Methodological gap, research intuition, and complete proposal.

IdeaScientist, arXiv:2610.04074 (2026)

IdeaScientist: RL in hypothesis generation

Reinforcement Learning

- A separate LoRA adapter for each role; asynchronous GRPO with role-specific rubric rewards.

$$r_{\text{rubric}} = g\left(\lambda_{\text{free}} r_{\text{free}} + \lambda_{\text{ref}} r_{\text{ref}} + \lambda_{\text{cit}} r_{\text{cit}}\right)$$

$$\boxed{r = 0.85 r_{\text{rubric}} + 0.15 r_{\text{proc}}}$$

- **Reference-free:** Specificity, significance, feasibility, and evidence.
- **Reference-grounded:** Agreement on substantive research decisions.
- **Citation F1:** Overlap of cited papers with the reference.

IdeaScientist, arXiv:2610.04074 (2026)

Other papers: complementary capabilities

System / publication date	Contribution and paper link
The AI Scientist Sakana AI et al. 25 March 2026	Agentic code-tree search within an end-to-end research workflow. Nature: 10.1038/s41586-026-10265-5
Paper2Agent Stanford 16 September 2026	Converts papers, code, and data into tested MCP tools and interactive agents. Nature: 10.1038/s41586-026-11044-y
ERA Google 19 May 2026	LLM code revision and tree search optimize empirical scientific software. Nature: 10.1038/s41586-026-10658-6
AI X-ray Scientist SLAC / Stanford et al. 1 July 2026	Tool-using agent for crystal alignment: simulated beamline to real instrument, with multimodal observation feedback. NMI: 10.1038/s42256-026-01261-5

Reference

Part II: ML in Nuclear Astrophysics

Exp Optimization: Online Bayesian Optimization

Scientific question

Can beam-based Bayesian optimization align SECAR and improve mass-plane optics with fewer measurements than manual tuning?

ML problem: Sample-efficient optimization of measured beam metrics.

Mathematical Abstraction

Alignment

$$\mathbf{u}_s^* = \arg\min_{\mathbf{u}_s} d_{\mathrm{FP1}}(\mathbf{u}_s)$$

Mass-plane optics

$$f_{\mathrm{FP2}}(\mathbf{u}_m) = \left(\frac{x + w_x - x_0}{20}\right)^4 + \left(\frac{w_y}{70}\right)^4$$

$\mathbf{u}_s, \mathbf{u}_m$: steerer and multipole settings; d_{FP1}: spot-center spread.

x_0: beam axis; w_x, w_y: spot widths.

Miskovich et al., PRAB 25, 044601 (2022)

Online Bayesian Optimization: Method and Results

Technical Stack

- **Diagnostics:** FP1 spot centers; FP2 position and widths.
- **Surrogate:** A Gaussian process, updated after each measurement.
- **Next setting:** Minimize $\mathrm{LCB}(\mathbf{u}) = \mu(\mathbf{u}) - \xi\sigma(\mathbf{u})$.
- **Control:** GPy / GPyOpt + PyEpics; steerers, then multipoles.

Results

- Angular deviation < 1 mrad.
- At least $3\times$ faster than manual tuning.
- 32% smaller beam-spot width for ^{133}Cs.

Miskovich et al., PRAB 25, 044601 (2022)

Exp Optimization: SECAR (p, n) Optics

Scientific question

Which SECAR settings separate nearly isobaric (p, n) recoils from the beam while retaining enough recoils for a cross-section measurement?

ML problem: Simulation-guided multi-objective optics optimization.

Mathematical Abstraction

$$\mathbf{u} \in \mathbb{R}^{19} \xrightarrow{\text{COSY-INFINITY}} (\Delta_{\mathrm{FP2}}, \Delta_{\mathrm{FP3}}, w_1, \ldots, w_K)$$

$$\text{maximize} \quad (\Delta_{\mathrm{FP2}}, \Delta_{\mathrm{FP3}})$$

$$\text{minimize} \quad (w_1, \ldots, w_K)$$

Δ: recoil–beam separation at slits; w: beam width along the separator.

Tsintari et al., PRR 7, 013074 (2025)

SECAR (p, n) Optics: Method and Results

Technical Stack

- **Search:** 19 field settings; recoil separation and beam widths.
- **Simulation:** COSY-INFINITY beam and recoil transport.
- **Optimization:** MOEA/D, Tchebycheff decomposition, differential evolution, and polynomial mutation.
- **Validation:** Beam tests of Pareto-selected optics; GP transmission interpolation.

Results

New ion optics enabled an astrophysical (p, n) measurement:

$$^{58}\mathrm{Fe}(p, n)^{58}\mathrm{Co}, \qquad \boxed{\sigma = 20.3 \pm 6.3 \text{ mb}}.$$

Tsintari et al., PRR 7, 013074 (2025)

Deepfield

An AI-driven Nuclear Astrophysics Laboratory

Name: Why Deepfield?

- **Deep → nuclear; Field → astrophysics.**
- **Deep / Field → depth / ground:**
  Jinping's deep underground laboratory.
- **Hubble Deep Field (HDF).**

The Hubble Deep Field experiment

In 1995, Robert Williams proposed observing an apparently empty patch of sky:

What would we discover if we observed it for long enough?

Ten days of observations in Ursa Major revealed **about 3,000 galaxies**.

Significance: From studying known targets to exploring unknown regions—how do search strategies and resource allocation shape discovery?

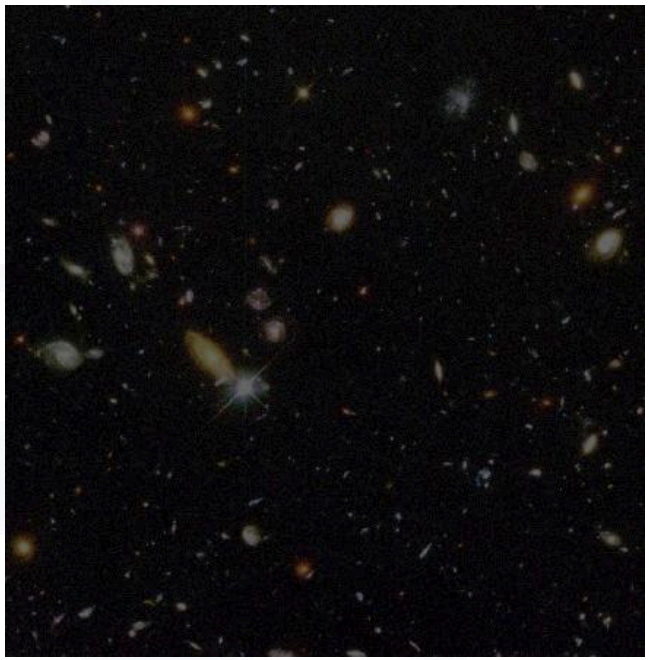


Figure 6: Central portion of the Hubble Deep Field, observed in 1995.

NASA, Robert Williams, and the Hubble Deep Field Team (STScI).

NASA Science: Hubble's Deep Fields

Deepfield Research Proposal

An AI-Driven, Human-in-the-Loop Laboratory

Setting: JUNA / Jinping deep underground nuclear astrophysics experiments.

Research Vision

1. **Autonomous research:** Investigate literature, generate hypotheses, reuse and develop code, and design and analyze experiments.
2. **Human guidance:** Scientists revise goals, contribute expertise, resolve disagreements, and judge key decisions.
3. **Recursive improvement:** Use experimental feedback and reflection to improve knowledge, code, strategies, and model capabilities.
4. **Scientific loop:** Connect real cross-section measurements with Code-proposed observations, archival analyses, or targeted simulations.
5. **Long-term aim:** Accumulate research experience, proactively discover scientific questions, and improve human–AI discovery capabilities.

Acceptance Criteria

For one scientist-confirmed reaction question

1. Identify evidence-grounded energy regions and competing hypotheses.
2. Complete a physical cross-section measurement and reproducible analysis.
3. Have Code proactively propose a discriminating observation, independent archival analysis, or targeted simulation.
4. Complete the follow-up with the responsible teams and independently verify its results.
5. Compare astrophysical conclusions before and after measurement.

Replay supports development; prospective acceptance needs a physical measurement. Unexecuted observing applications remain pending. Report observations and simulations by their actual evidence type.

Research Loop

One Loop connects two experiment types

1. **Literature:** Identify reactions, energy regions, and evidence under scientist-defined goals and constraints.
2. **Hypothesis + Code:** Develop competing mechanisms and quantitative predictions using verified, reusable tools.
3. **First experiment:** Experiment Agent and the human team optimize and perform a real cross-section measurement.
4. **Code-led follow-up:** Infer cross sections and rates; propose discriminating observations, archival analyses, or simulations.
5. **Execution:** Code runs simulations and analyzes data; Experiment coordinates new observations with scientists and observing teams.
6. **Interpretation and next cycle:** Independently verify results; scientists judge explanations and update evidence, models, and questions.

Robin, Nature (2026)

Core MAS Roles

Role	Main responsibility
Literature	Evidence, conflicts, knowledge gaps, reactions and energy regions; reusable data, code, and methods.
Hypothesis	Generate, debate, refine, and select competing mechanisms; predictions, falsification, and feedback from both experiment types.
Code	Reuse, adapt, or develop tools; infer cross sections and rates; run astrophysical simulations and proactively propose follow-ups.
Experiment	Feasible designs, beam measurements, and instrument feedback; coordinate new observations with scientists and observing teams.
Verification	Independent evidence, physical and numerical checks; uncertainty, reproducibility, and external validation.
Supervisor	Tasks, shared state, memory, dependencies, resources, and recovery.

Scientists set and revise goals and judge scientific conclusions. Research directions define capabilities; Agent roles define responsibilities.

Candidate Agents for Hypothesis Refinement

Candidate role	Responsibility
Reflection	Check assumptions, evidence, counterexamples, and testability.
Ranking	Pairwise comparisons of scientific value, feasibility, and resource needs; retain disagreements.
Proximity	Cluster and de-duplicate hypotheses; identify complementary mechanisms and unexplored directions.
Evolution	Revise or combine hypotheses; retain versions and reasons for changes.
Review synthesis	Feed recurring errors and knowledge gaps into later generation.

Begin as Hypothesis Agent subtasks; split roles only when evaluation justifies the cost.

Internal rankings are not scientific evidence.

Towards an AI co-scientist, arXiv:2502.18864

Data Engineering

- **Acquire:** Papers, formulas, tables, curves, nuclear data, experiments, simulations, and observations.
- **Normalize:** Identifiers, units, energy frames, metadata, and validity ranges.
- **Align:** Connect modalities by reaction, energy region, conditions, and observables.
- **Preserve provenance:** Originals, versions, calibration, uncertainty, covariance, and conflicts.
- **Build datasets:** Expert-reviewed examples; independent splits by source, time, or reaction.

Harness Engineering

- **Orchestrate:** Supervisor-managed dependencies, handoffs, priorities, and budgets.
- **Execute:** Parallel independent work; respect measurement and analysis dependencies.
- **Recover:** Track runs, checkpoints, failures, and affected outputs before resubmission.
- **Workspace:** Natural-language goals and feedback; inspect hypothesis branches, evidence, data, code, and research progress.
- **Evaluate:** Record traces, costs, and outputs; compare workflows independently.

Towards an AI co-scientist, arXiv:2502.18864

Prompt Engineering

- **Roles:** Define scientific responsibility, tools, and collaboration boundaries.
- **Tasks:** State questions, assumptions, units, reference frames, validity, and outputs.
- **Evidence:** Require competing explanations, predictions, uncertainty, and falsification criteria.
- **Code priority:** Search, reproduce, and reuse before adapting or developing.
- **Feedback:** Version prompts; evaluate changes against scientist feedback and diagnostics.

Towards an AI co-scientist, arXiv:2502.18864; ERA, Nature (2026)

Context Engineering

- **Working context:** Current question, constraints, evidence, hypotheses, tool feedback, and next actions.
- **Persistent memory:** Separate shared task state, long-term evidence and experience, and complete research artifacts.
- **Retrieve and compress:** Preserve numbers, units, correlated errors, counterevidence, and disagreements.
- **Version:** Track provenance, revisions, validation status, and obsolete conclusions.
- **Handoff and resume:** Share task IDs and artifacts; recover valid state from checkpoints.

Towards an AI co-scientist, arXiv:2502.18864

Tool Use

Invocation loop

Scientific purpose → tool selection → input checks → execution → numerical and physical checks → state update.

- **Library:** Literature and nuclear data, scientific software, code execution, analysis, instruments.
- **Reuse:** Reproduce paper methods before exposing MCP tools, APIs, or wrappers.
- **Contracts:** Versions, formats, units, frames, validity, costs, and invocation conditions.
- **Feedback:** Diagnose failures; distinguish execution success, numerical validity, and scientific evidence.

Paper2Agent, Nature (2026); ERA, Nature (2026)

Multimodal Fusion and Model Training

- **Align evidence:** Match literature, nuclear data, spectra, energy levels and reaction networks, simulations, and observations by reaction, energy, and environment.
- **Reason and quantify:** Combine mechanistic reasoning with numerical predictions and uncertainty to evaluate competing hypotheses.
- **Establish baselines:** First test existing models, retrieval, and scientific tools; compare separate and combined use.
- **Explore training:** With expert-reviewed samples, study domain continual pretraining, SFT, and role-specific LoRA / reinforcement learning.
- **Evaluate contributions:** Separate knowledge updates, prompt/code strategies, and parameter training; distinguish LLM confidence from physical uncertainty.

XunZi, Nature Biomedical Engineering (2026); IdeaScientist, arXiv:2610.04074 (2026)

Validation

- **Software:** Reference reproduction, convergence, and executable regression cases.
- **Physics:** Units, dimensions, scales, conservation, positivity, and model validity.
- **Uncertainty:** Calibration, interval coverage, and stability across reasonable analyses.
- **Evidence:** Independent checks against competing predictions; distinguish simulations from observations.
- **System:** Re-evaluate changes; track outcomes, failures, human input, and resource costs.

Six Research Directions

Direction	Research focus
I. Knowledge and discovery	Evidence, predictive models, experience, and long-term research state.
II. Hypotheses	Doubt, cross-domain and open-world generation, robustness, and selection.
III. Analysis and simulation	Code reuse and development, measurement analysis, simulation, and follow-ups.
IV. Experiment design	Uncertainty-aware, information-driven design and optimization.
V. Experimentation	Beam, astrophysical, and numerical experiments.
VI. Benchmarking	Literature, hypothesis, code, and experiment evaluation.

Near term: a complete JUNA Loop. Longer term: transfer across reactions, environments, and scientific fields.

I. Continual Knowledge-Guided Exploration

- **Continual evidence alignment:** Integrate evolving literature, nuclear data, and experiments; retain provenance, validity, conflicts, and candidate reactions or energy regions.
- **Scientific world models:** Discover and revise models that make quantitative predictions, using physically constrained structure, equation, or program search; Direction III implements and tests them.
- **Experimental experience:** Link configurations, raw data, analyses, anomalies, and successful or failed procedures to hypotheses and physical models.
- **Long-term research state:** Maintain tasks, dependencies, hypothesis branches, pending experiments, and recovery across weeks or months.

Evidence records, predictive models, experimental experience, and task state serve different roles. Test whether continual evidence alignment improves quantitative research decisions.

II. Hierarchical Scientific Hypothesis Generation

- **Hierarchical doubt:** Question assumptions, approximations, validity, and anomalies across measurements, nuclei, networks, astrophysics, and fundamental physics.
- **Cross-domain reasoning:** Transfer theories, mathematical structures, and mechanisms through analogy and abduction.
- **Simulation-guided discovery:** Use Code scans, sensitivity, and uncertainty to reveal dominant factors, model mismatch, and new questions.
- **Mechanisms:** Derive competing causal explanations, observable predictions, and their uncertainty.
- **Open-world generation:** Explore unlisted variables, mechanisms, and questions; request new or revised models in Direction I.
- **Robustness and falsifiability:** Refine using both experiment types; test parameter, model, and uncertainty assumptions and state falsification conditions.
- **Quantitative selection:** Combine scientific value, novelty, information gain, sensitivity, uncertainty, and feasibility; retain complementary hypotheses.

Output: Versioned hypothesis cards with evidence, predictions, falsification tests, and revisions.

III. Data Analysis & Simulation with Codes

Reuse first

Search → reproduce → check validity → reuse → adapt or extend → fill remaining gaps → validate.

- Reproduce and reuse reaction, network, stellar, and analysis software; verify reference results and validity.
- Adapt or extend first; develop new code only for unmet needs, guided by execution feedback and numerical / physical checks.
- Infer cross sections and rates; check calibration, background, efficiency, and statistical / systematic uncertainty.
- After analysis, **proactively propose discriminating observations, archival analyses, or targeted simulations**.
- Run astrophysical simulations and analyze observations; archive code, environments, provenance, figures, and research reports.

ERA, Nature (2026)

IV. Experimental Design & Optimization

- **Formulate:** Model observables, competing hypotheses, uncertainty, and constraints; target information gain or parameter precision.
- **Design offline:** Use Bayesian design and multi-objective optimization for energies, beam, target, detectors, time, and resources.
- **Simulate:** Background, efficiency, response, feasibility, and objective trade-offs.
- **Adapt online:** Bayesian updates and optimization using measurements and scientist feedback.
- **Prepare execution:** Validate interfaces in simulation or replay; assess Code-proposed follow-ups and coordinate observations with scientists and observing teams.

Output: Executable plans, configurations, calibration, and adjustment strategies; Direction V executes them.

Miskovich et al., PRAB 25, 044601 (2022); Tsintari et al., PRR 7, 013074 (2025)

V. Experimentation Implementation

- **Beam experiments:** Experiment Agent and the human team perform real measurements, calibration, background studies, and repetitions.
- **Astrophysical experiments:** Carry out Code-proposed observations with scientists and observing teams, or analyze independent archives to test nuclear-constrained predictions.
- **Numerical experiments:** Code Agent reuses or extends reaction and stellar codes for controlled variations, competing and counterfactual models, and uncertainty propagation.

Beam measurement is the first experiment type; the post-measurement observation or simulation is the second.

Preserve records and negative results; return supported, excluded, and unresolved explanations to the next cycle. Distinguish simulation deductions from independent observational evidence.

VI. Benchmark

Benchmark	Main evaluation targets
Literature	Retrieval, figures, formulas, nuclear data, provenance, conflicts, and energy-region evidence.
Hypotheses	Novelty, plausibility, mechanisms, testability, falsifiability, and evidence-driven revision.
Code	Discovery, reproduction, reuse, adaptation, necessary development, numerical / physical validity, reproducibility, uncertainty, and cost.
Experiments	Feasibility, optimization, feedback, reliability, information gain, and astrophysical interpretation.

Evaluation basis

Use original data and expert references. Check reproducibility and evaluate whether follow-ups distinguish mechanisms and improve astrophysical explanations.

Independent Evaluation and Ablation

- **Independent tests:** Hold out by publication time, reaction system, or experimental batch; check overlap.
- **Baselines:** Human workflows, baseline Agents, full MAS; match resource budgets.
- **Ablations:** Tools, memory, fusion, and candidate roles; include coordination cost.
- **Key comparison:** Measurement and analysis alone versus the full Loop with proactive follow-ups.
- **Report:** Physics checks, calibrated uncertainty, evidence, human effort, and costs.

Separate replay, simulation, and prospective physical results. Internal rankings and analysis consensus are not independent evidence.

Example Scientific Loop

Illustrative workflow; not an established discovery

1. **Literature:** Find a reaction, key energy regions, and conflicting evidence.
2. **Hypothesis:** Propose competing resonance mechanisms.
3. **Code:** Reproduce calculations and predict cross sections and astrophysical consequences.
4. **Experiment:** Measure at JUNA; extract cross sections, rates, and uncertainty.
5. **Follow-up:** Code proposes a discriminating test; run targeted simulations or analyze new / archival observations after coordinated execution.
6. **Interpretation:** Check results independently; return new constraints and unresolved questions to the next cycle.

Seven-Month Research Roadmap

Month	Focus and milestone
1	**Literature:** Evidence, confirmed pilot, and beam-time coordination.
2	**Code:** Reproduce at least one reference within expert tolerances; reusable tools and a discriminating follow-up proposal.
3	**Hypothesis:** At least two distinguishable explanations; quantitative predictions and falsification criteria.
4	**Experiment:** Approved plan; optimization and instrument-feedback demo.
5	**Beam:** Real cross sections and rates; statistical and correlated systematic uncertainties.
6	**Follow-up:** Execute the second experiment; compare astrophysical conclusions before and after measurement.
7	**Integration:** Frozen independent benchmarks, matched-budget baselines and ablations, research report.

Develop Code tools before Hypothesis; the scientific Loop stays iterative. Shared engineering throughout; monthly benchmarks and frozen versions. Beam scheduling from Month 1. Replay tests capability; delayed real measurements leave the Loop pending.

Seven-Month Research Budget

CNY 100,000; single-developer project

Item	CNY
Large language model APIs	40,000
CPU and GPU computing	25,000
Expert review, annotation, verification	20,000
Retrieval, OCR, embeddings	5,000
Contingency	10,000
Total	**100,000**

Expert services are combined; CPU and GPU resources are used on demand.

Excludes developer compensation, beam experiments, experiment-related travel, and storage.

Thank You

Questions and Discussion

References: ML in Nuclear Physics

[1] R. Lasseri, D. Regnier, J. Ebran, and A. Penon. Taming Nuclear Complexity with a Committee of Multilayer Neural Networks. *Phys. Rev. Lett.* **124**, 162502 (2020).

[2] R. Barsotti and M. R. Shepherd. Using machine learning to separate hadronic and electromagnetic interactions in the GlueX forward calorimeter. *JINST* **15**, P05021 (2020).

[3] C. Fanelli and J. Pomponi. DeepRICH: Learning deeply Cherenkov detectors. *Mach. Learn.: Sci. Technol.* **1**, 015010 (2020).

[4] A. Scheinker et al. Model-independent tuning for maximizing free electron laser pulse energy. *Phys. Rev. Accel. Beams* **22**, 082802 (2019).

References: ML in Astrophysics

[5] K. Moriwaki, T. Nishimichi, and N. Yoshida. Machine learning for observational cosmology. *Rep. Prog. Phys.* **86**, 076901 (2023).

[6] T. S. Fraser, R. Tojeiro, and H. G. Chittenden. Applying unsupervised learning to resolve evolutionary histories and explore the galaxy-halo connection in IllustrisTNG. *MNRAS* **522**, 5758–5774 (2023).

[7] R. Arimura. A machine learning approach for mass composition analysis with TALE-SD data. *EPJ Web Conf.* **283**, 02011 (2023).

[8] J. W. den Hartogh et al. Barium stars as tracers of s-process nucleosynthesis in AGB stars. II. Using machine learning techniques on 169 stars. *Astron. Astrophys.* **672**, A143 (2023).

[9] M. S. Smith and D. Lu. Machine learning opportunities for nucleosynthesis studies. *Front. Astron. Space Sci.* **11**, 1494439 (2024).

References: Scientific Reasoning and Agents

[10] A. E. Ghareeb et al. A multi-agent system for automating scientific discovery (Robin). *Nature* (2026). DOI: 10.1038/s41586-026-10652-y.

[11] J. Gottweis et al. Towards an AI co-scientist. arXiv:2502.18864 (2025). Journal version: Accelerating scientific discovery with Co-Scientist, *Nature* (2026).

[12] X. Huang et al. XunZi, an AI biologist, reveals disease-modifying targets. *Nature Biomedical Engineering* (2026). DOI: 10.1038/s41551-026-01769-6.

[13] J. Liu et al. IdeaScientist: Orchestrating Agents for Grounded Scientific Ideation. arXiv:2610.04074 (2026).

References: Code and Experimental Workflows

[14] C. Lu et al. Towards end-to-end automation of AI research. *Nature* **651**, 914–919 (2026).

[15] J. Miao, J. R. Davis, Y. Zhang, J. K. Pritchard, and J. Zou. Reimagining research papers as interactive and reliable AI agents. *Nature* (2026). DOI: 10.1038/s41586-026-11044-y.

[16] E. Aygün et al. An AI system to help scientists write expert-level empirical software. *Nature* (2026). DOI: 10.1038/s41586-026-10658-6.

[17] Z. Chen et al. An agentic artificially intelligent X-ray scientist. *Nature Machine Intelligence* **8**, 1075–1086 (2026).

References: Nuclear Experiments and Deep Field

[18] S. A. Miskovich et al. Online Bayesian optimization for the SECAR recoil mass separator. *Phys. Rev. Accel. Beams* **25**, 044601 (2022).

[19] P. Tsintari et al. Machine learning enabled measurements of astrophysical (p, n) reactions with the SECAR recoil separator. *Phys. Rev. Research* **7**, 013074 (2025).

[20] NASA Science. Hubble's Deep Fields. Hubble Space Telescope science resource; source of the Deep Field image and observing history.